P1-14 (companion) — Proposed Manuscript Structure and Readiness Map

Command: P1-14 — Mathematical Advancement Without L1 (companion deliverable) **Date:** 21 October 2026

Status: PROPOSAL ONLY. This is a structure plan, not a manuscript. No text below asserts novelty or priority. Gate G1 (literature/novelty) remains **DEFERRED**; manuscript drafting stays gated until the Commander decides otherwise (repo L1 full text still unacquired — see P1_12_L1_ACQUISITION_STATUS.md). Mathematical readiness tags come from P1_14_MATHEMATICAL_ADVANCEMENT.md (proofs: P1-04…P1-06 and P1-14 §4; computational evidence: code/validate_p1_14.py).

Readiness tags: READY = proof complete and independently audited · PROPOSED = proof complete in P1-14, needs Commander adoption decision before it may be listed as a paper result · SUPPORTING = computational-evidence section material · BLOCKED = cannot be written honestly yet · ⚠ CONDITIONAL = only under extra external hypotheses.

Proposed section structure (aligned with the frozen blueprint §4 architecture)

1. Introduction — BLOCKED (Gate G1)

- **May be written now (internal draft only):** the problem statement (joint (dim C, dim Hull) distribution for labeled repeated-root constacyclic codes over A ≅ ∏ K_s), the model, and a neutral summary of contributions phrased **without** "first / new / novel" priority language.
- **May not be written yet:** any priority positioning against repo L1 (Debnath–Prakash–Islam 2023 — unread at theorem level), any "to our knowledge the first" sentence, and any related-work table that ranks our results against L1/L4/L8 beyond what P1-09 verified from actual texts. The honest-distance paragraphs vs L4 (arXiv:2412.08512) and L8 (Eldin & Solé, arXiv:2305.15237) can be drafted from P1_09_OVERLAP_MATRIX.md, with L1 explicitly marked "comparison pending full text".

2. Preliminaries and conventions — READY

Notation freeze; code-first pairing (S7) stated verbatim (it is load-bearing for T3's α_{i−1}); # with the f(0) ≠ 0 domain (PC-3); compatibility λ_s^{1+p^{d_s−k}} = 1; labeled assignments over fixed labeled components (PC-5). Sources: blueprint §0 + P1-04 §0. Include the orientation warning (P1-14 §2.3): the predecessor convention must be visible wherever T3/T4 are stated.

3. Repeated-root factorization and parametrization — READY

M_s = S_s^P; T1 (bijection, (P+1)^{|F_s|}); T2 (# multiplicativity with multiplicities, root transport, compatibility equivalence). Proofs: P1-04 §§1–2 (RR-1, RR-2). No separability is used in RR-1/RR-3 — worth one remark each, as referees will ask.

4. The Galois dual — READY

Euclidean-dual lemma RR-3 (no separability anywhere; the identity gh = x^n − λ does all the work); transport D(C) = ρ(C^⊥); T3 with β_i = P − α_{i−1}; the remark that the generator identity survives without compatibility in the two-modulus form x^n − λ^{−p^{d_s−k}}, while the same-modulus statement needs compatibility (necessity witness: P1-04 §7, GF(9) order-8 λ).

5. Hull generator and dimension — READY

RR-5 (intersection = lcm, in K[x] and in the quotient); T4 with the edge term min(P − α_i, α_{i−1}). Proof: P1-04 §6. This is the paper's computational heart; the P − max(u,v) = min(P−u, P−v) step should be displayed, not hidden.

6. Transfer matrix and the global enumerator — READY

T5 (closed-walk trace; labeled positions — no division by L); CRT/product assembly (Prop. 3.1 — direct product of the square-free algebra, no coprimality between components needed); T6 (main theorem). Proofs: P1-05 §§2–4. Include the hypothesis-separation remark (P1-14 §2.2): the trace identity is pure Z[u,z] combinatorics valid for all P ≥ 1; the prime-power enters through T1/T4.

7. Corollaries: counts, LCD, self-orthogonality — READY (+ one refinement)

- C1 (total), C2 (code-dimension marginal) — .
- C3 (LCD 2^{#orbits}, P-independent) — . **N3 refinement** (joint LCD × code-dimension generating function ∏_O (1 + u^{P w_O L_O})): proved in P1-14 §4; include only after adoption decision.
- C4 (self-orthogonal ⟺ α_{i−1}+α_i ≥ P per edge; dual-containing ⟺ ≤ P; both counts, equal by α ↦ P−α) — (PC-4 text already in the frozen blueprint).

8. Moments: mean, variance, and a covariance theorem — READY (+ two additions)

- C5 corrected mean (ν_P on L_O=1, μ_P on L_O≥2) and the L=1 self-loop phenomenon — (frozen corrected text; P1-06 Thm. 2.4). The pre-correction failure should be footnoted honestly if the predecessor manuscript used the unsplit form (P1-14 §5.3 gives two quantified witnesses).
- C6 variance (three regimes V_1/V_2/V_{L≥3}; weights w_O vs w_O²) — (frozen text; P1-06 Thm. 3.7; symbols re-derived in P1-14 Layer A).
- **N1** Cov(dim C, dim Hull) = 0 with the uncorrelated-≠-independent witness — proved in P1-14 §4; adoption decision pending. If adopted, it belongs here as a small theorem (its proof is short and the statement is quotable).
- **N2** Var(dim C) = [P(P+2)/12]·Σ w_f² — proved in P1-14 §4; adoption decision pending; natural companion to N1 (together they give the full second-order summary of the joint distribution's margins).

9. Examples and computational validation — SUPPORTING (ready to assemble)

Material exists and is independently reproduced by code/validate_p1_14.py: E1 (P=2,L=2 nine-tuple table — suitable as a hand-checkable first example), E2 (P=1,L=3 simple-root recovery), E3 (GF(4), n=10, mixed orbits [1,2]), E4 (GF(8), n=14, orbit length 6 — also the orientation witness), E5 (two-component GF(2)×GF(4) assembly, 729 assignments). The section must carry the standing disclaimer: finite checks are supporting evidence, not proof. Suggested: print full joint distributions for E3/E4 (they are small) and the E1 table.

10. Limitations and scope boundaries — READY (content fixed; wording at drafting time)

Labeled (non-isometry) counting; compatible twists only; square-free affine A; uniform labeled model for moments; orientation convention borne by T3 (P1-14 §2.3); incompatible-twist two-modulus form of T3 as the only statement available there; Cov = 0 is not independence (N1 remark).

11. Conditional quantum remark (C7) — ⚠ CONDITIONAL / recommend appendix or omission

Only in the finite-field case with an externally known distance d (standard [[n, K−H, d; n−K−H]] form). The enumerator alone yields no distance or quantum parameter. Recommendation: omit from the main text, or one clearly-fenced remark; never a numbered consequence of T6.

12. Conclusion — BLOCKED with §1

Depends on the introduction's final positioning; draft only after Gate G1 is decided.

Readiness summary

Block | Content | Tag

§2–§6 | Conventions, T1–T6 | READY (proved; independently audited in P1-14)

§7 | C1–C4 | READY

§7/§8 | N3, N1, N2 | PROPOSED — proved in P1-14 §4; Commander adoption decision required (frozen blueprint untouched until then)

§8 | C5 (corrected), C6 | READY (frozen corrected text; symbolically re-derived)

§9 | Examples/validation E1–E5 | SUPPORTING — data reproduced by fresh import-free code

§10 | Limitations | READY (content fixed)

§11 | C7 EAQECC | ⚠ CONDITIONAL — appendix or omit

§1, §12 | Introduction positioning, conclusion | BLOCKED until Gate G1 (L1 theorem-level comparison)

What this structure deliberately does not do: claim novelty or priority for any block; rank the results against repo L1; present computational checks as proofs; or fold N1–N3 into the theorem list before adoption. The mathematical core (§§2–10) is, as of P1-14, in a state where drafting could begin the day the Commander (i) decides on N1–N3 adoption and (ii) resolves Gate G1 — those are the only two gates left, and the second one still runs through the lawful acquisition of L1 (P1_11_L1_ACQUISITION_HANDOFF.md, drafts **NOT SENT**).